Guwahati, Assam: regional check

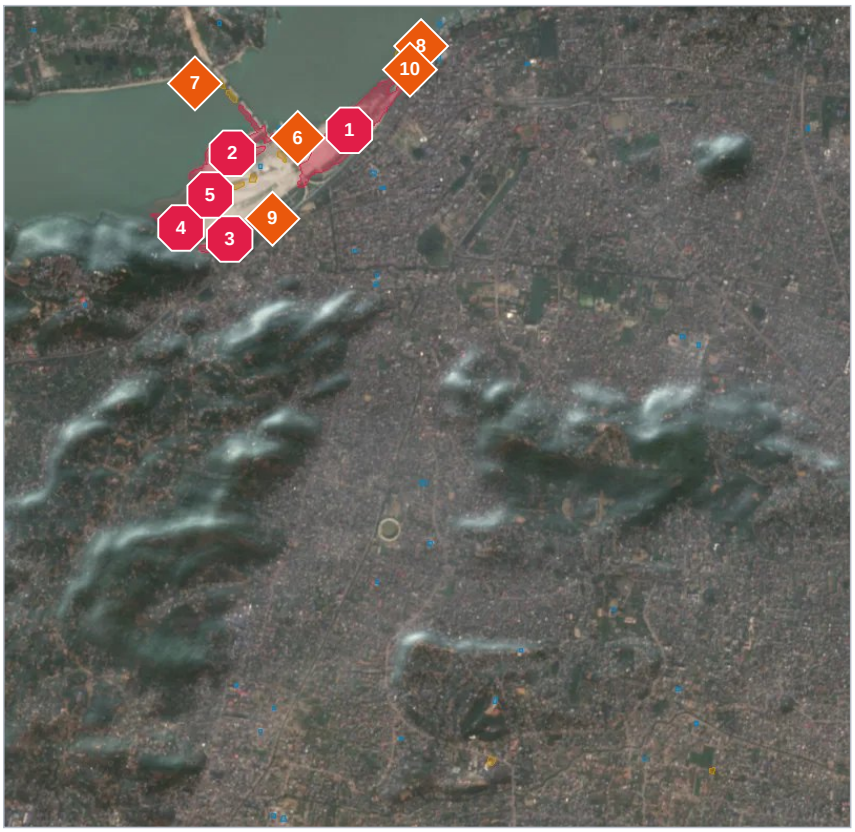
Guwahati, Assam | Before - Sentinel-2 - 2019-01-12 to After - Sentinel-2 - 2025-01-15

What we found

We found 51 real changes covering 74.64 hectares. The biggest and strongest (rank 1) is water receded: 29.59 ha at 26.1799, 91.7324.

By priority: 5 Critical, 5 High, 9 Medium, 32 Low.

The detector first flagged 4,932 candidate areas. The rest were removed as false alarms: seasonal crop and vegetation cycles, change types this analysis does not track, tiny specks and noise, cloud and cloud shadow, areas below the minimum size.



Numbers = rank of the 10 highest-priority changes. Outline colour = priority.

Changes by priority

Critical	5
High	5
Medium	9
Low	32

Totals

Area changed: 74.64 ha

Cropland affected: 4.62 ha (upper bound)

Built-up land affected: 1.89 ha

How to read this report

Rank	Every change has a number. Rank 1 is the highest priority: the largest and strongest change. Priority = size x strength of the change.
Priority colour and shape	Critical (octagon): score 20,000 or more (about 2 ha of strong change); High (diamond): score 5,000 or more; Medium (square): score 1,500 or more; Low (circle): score below 1,500 (small or weak).
Confidence	How sure the system is that the change is real, from 0 to 100 %. Changes marked 'needs a human check' sit between 50 % and 70 %.
What changed	New construction, vegetation cleared, vegetation growth, new water / flooding, water receded, or unexplained change. These come from rules on the satellite colours, not from knowing what is actually on the ground.
Raw detections	Everything the detector first flagged. Most are not real (clouds, crop seasons, shadows, specks), so only changes that pass the filters are listed here.

The 10 highest-priority changes

Start here. Each card says where the change is, how big it is, why it was flagged and what to do.

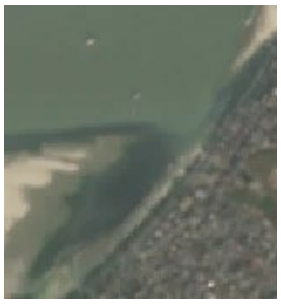
1

Water receded

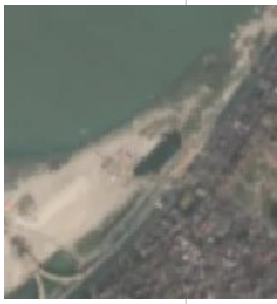
Rank 1 of 51 | Critical priority

Size	29.59 ha (295,850 m2), about 41 football pitches
Confidence	99 %
Location	26.17991, 91.73243 (open in Google Maps)
What it is	Water that was there before has gone.
Why flagged	built-up index rose (+0.47), vegetation index rose (+0.19), water index fell (-0.28), SWIR/blue ratio (paved surfaces are flatter) rose (+1.32)
What to do	Check first. A large, strong change; send someone or review recent imagery now.

Before



After



2

Water receded

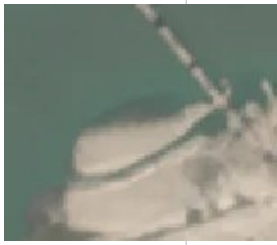
Rank 2 of 51 | Critical priority

Size	17.97 ha (179,700 m2), about 25 football pitches
Confidence	99 %
Location	26.17791, 91.72117 (open in Google Maps)
What it is	Water that was there before has gone.
Why flagged	built-up index rose (+0.54), vegetation index rose (+0.28), water index fell (-0.38), SWIR/blue ratio (paved surfaces are flatter) rose (+1.30)
What to do	Check first. A large, strong change; send someone or review recent imagery now.

Before



After



3

Water receded

Rank 3 of 51 | Critical priority

Size	5.68 ha (56,800 m2), about 8 football pitches
Confidence	98 %
Location	26.17047, 91.72092 (open in Google Maps)
What it is	Water that was there before has gone.
Why flagged	built-up index rose (+0.16), vegetation index rose (+0.26), water index fell (-0.32), SWIR/blue ratio (paved surfaces are flatter) rose (+1.03)
What to do	Check first. A large, strong change; send someone or review recent imagery now.

Before



After



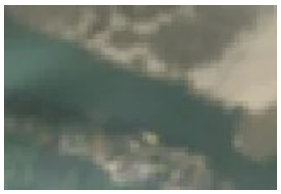
4

Water receded

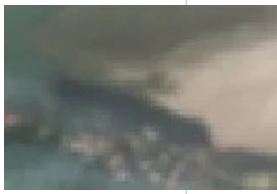
Rank 4 of 51 | Critical priority

Size	3.76 ha (37,600 m2), about 5 football pitches
Confidence	98 %
Location	26.17143, 91.71623 (open in Google Maps)
What it is	Water that was there before has gone.
Why flagged	built-up index rose (+0.28), vegetation index rose (+0.26), water index fell (-0.38), SWIR/blue ratio (paved surfaces are flatter) rose (+1.32)
What to do	Check first. A large, strong change; send someone or review recent imagery now.

Before



After



5

New construction

Rank 5 of 51 | Critical priority

Size	2.58 ha (25,800 m2), about 4 football pitches
Confidence	98 %
Location	26.17429, 91.71904 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.13), vegetation index rose (+0.06), SWIR/blue ratio (paved surfaces are flatter) fell (-0.38)
What to do	Check first. A large, strong change; send someone or review recent imagery now.

Before



After



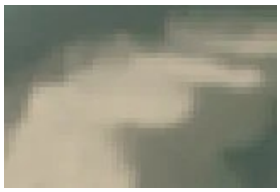
6

New construction

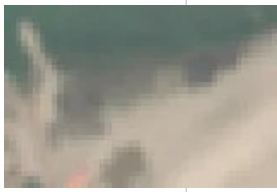
Rank 6 of 51 | High priority

Size	2.15 ha (21,500 m2), about 3 football pitches
Confidence	98 %
Location	26.17925, 91.72746 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.06), water index rose (+0.06), SWIR/blue ratio (paved surfaces are flatter) fell (-0.54)
What to do	Check soon. A substantial change that is probably real.

Before



After



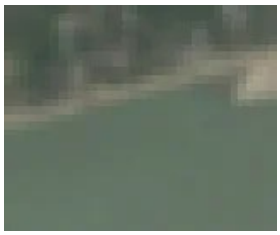
7

Water receded

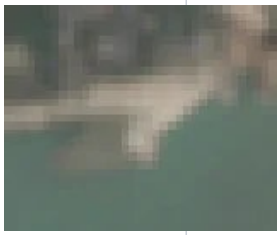
Rank 7 of 51 | High priority

Size	1.67 ha (16,650 m2), about 2 football pitches
Confidence	98 %
Location	26.18398, 91.71758 (open in Google Maps)
What it is	Water that was there before has gone.
Why flagged	built-up index rose (+0.41), vegetation index rose (+0.22), water index fell (-0.30), SWIR/blue ratio (paved surfaces are flatter) rose (+1.12)
What to do	Check soon. A substantial change that is probably real.

Before



After



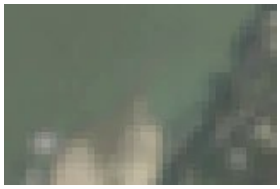
8

Water receded

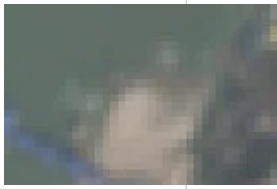
Rank 8 of 51 | High priority

Size	1.12 ha (11,250 m2), about 2 football pitches
Confidence	98 %
Location	26.18723, 91.73933 (open in Google Maps)
What it is	Water that was there before has gone.
Why flagged	built-up index rose (+0.31), vegetation index rose (+0.24), water index fell (-0.31), SWIR/blue ratio (paved surfaces are flatter) rose (+1.01)
What to do	Check soon. A substantial change that is probably real.

Before



After



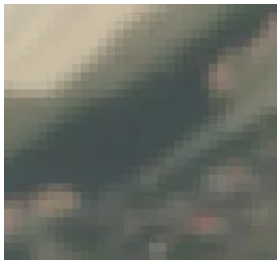
9

Water receded

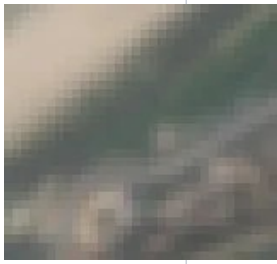
Rank 9 of 51 | High priority

Size	1.12 ha (11,200 m2), about 2 football pitches
Confidence	97 %
Location	26.17223, 91.72503 (open in Google Maps)
What it is	Water that was there before has gone.
Why flagged	built-up index rose (+0.09), vegetation index rose (+0.32), water index fell (-0.36), SWIR/blue ratio (paved surfaces are flatter) rose (+0.90)
What to do	Check soon. A substantial change that is probably real.

Before



After



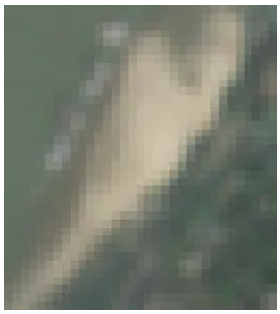
10

New construction

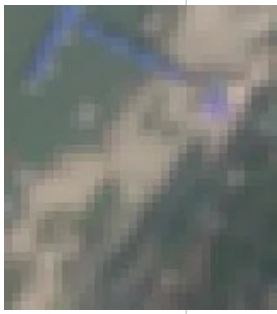
Rank 10 of 51 | High priority

Size	0.92 ha (9,200 m2)
Confidence	97 %
Location	26.18518, 91.73826 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.07), SWIR/blue ratio (paved surfaces are flatter) fell (-0.39)
What to do	Check soon. A substantial change that is probably real.

Before



After



All 51 changes found

Every change that survived the false-alarm filters, in priority order (rank 1 is the highest). A * after the confidence means it needs a human check. The CSV and GeoPackage downloads hold the same list with every attribute.

Rank	Priority	What changed	Size (ha)	Confidence	Latitude	Longitude
1	Critical	Water receded	29.59	99 %	26.17991	91.73243
2	Critical	Water receded	17.97	99 %	26.17791	91.72117
3	Critical	Water receded	5.68	98 %	26.17047	91.72092
4	Critical	Water receded	3.76	98 %	26.17143	91.71623
5	Critical	New construction	2.58	98 %	26.17429	91.71904
6	High	New construction	2.15	98 %	26.17925	91.72746
7	High	Water receded	1.67	98 %	26.18398	91.71758
8	High	Water receded	1.12	98 %	26.18723	91.73933
9	High	Water receded	1.12	97 %	26.17223	91.72503
10	High	New construction	0.92	97 %	26.18518	91.73826
11	Medium	Water receded	0.55	98 %	26.18393	91.72013
12	Medium	Water receded	0.51	98 %	26.18660	91.73722
13	Medium	Water receded	0.54	98 %	26.18277	91.72119
14	Medium	New construction	0.48	98 %	26.12516	91.74610
15	Medium	New construction	0.46	98 %	26.17502	91.72181
16	Medium	New construction	0.40	98 %	26.17564	91.72319
17	Medium	New construction	0.50	97 %	26.17757	91.72604
18	Medium	Water receded	0.25	99 %	26.12430	91.76735
19	Medium	New construction	0.26	97 %	26.18595	91.73810
20	Low	New construction	0.16	99 %	26.12038	91.72515
21	Low	New construction	0.20	98 %	26.18004	91.72584
22	Low	Water receded	0.14	99 %	26.16651	91.73498
23	Low	Water receded	0.26	95 %	26.14929	91.73955
24	Low	New construction	0.15	98 %	26.13142	91.76409
25	Low	New construction	0.20	96 %	26.12015	91.72610
26	Low	New construction	0.18	96 %	26.13041	91.74645
27	Low	New construction	0.16	97 %	26.16199	91.76442
28	Low	New construction	0.14	97 %	26.14400	91.74018
29	Low	New construction	0.15	97 %	26.18611	91.74110
30	Low	Water receded	0.15	97 %	26.18909	91.74112
31	Low	New construction	0.18	94 %	26.17224	91.72399
32	Low	New construction	0.18	94 %	26.17612	91.73480
33	Low	New construction	0.09	99 %	26.13472	91.74895
34	Low	New construction	0.12	97 %	26.17487	91.73567
35	Low	New construction	0.12	97 %	26.12841	91.76582
36	Low	New construction	0.12	97 %	26.18005	91.77648
37	Low	New construction	0.09	98 %	26.18565	91.74961
38	Low	New construction	0.12	97 %	26.16128	91.76598
39	Low	New construction	0.09	98 %	26.14065	91.73508
40	Low	New construction	0.09	98 %	26.16478	91.70701
41	Low	Water receded	0.09	98 %	26.12713	91.73743
42	Low	New construction	0.09	98 %	26.13177	91.72158
43	Low	New construction	0.09	98 %	26.18857	91.70215
44	Low	New construction	0.12	95 %	26.18913	91.71995
45	Low	New construction	0.12	94 %	26.17671	91.72389
46	Low	New construction	0.09	96 %	26.13832	91.75781
47	Low	New construction	0.09	96 %	26.12973	91.72520
48	Low	New construction	0.09	96 %	26.16943	91.73297

Rank	Priority	What changed	Size (ha)	Confidence	Latitude	Longitude
49	Low	New construction	0.09	96 %	26.16728	91.73510
50	Low	New construction	0.09	94 %	26.12534	91.76085
51	Low	New construction	0.09	94 %	26.12773	91.72392

Data sources

- Copernicus Sentinel-2 L2A (ESA), via the Element84 Earth Search COG bucket
- ESA WorldCover 2021 v200 (10 m), CC-BY 4.0

Limits - what this report cannot tell you

- Optical imagery cannot see through cloud; flagged areas with cloud are removed, not analysed.
- Change type comes from spectral index rules (no training); check critical changes on the ground.
- 10 m pixels cannot resolve single small buildings or people.
- Rivers, sandbars and lake edges move between dates and can be flagged as change; bare soil in dry regions can look like construction. Check flagged areas near water.
- Land cover is ESA WorldCover 2021, newer than the before image (2019): land already cleared or under construction can be mislabelled as cropland, so treat cropland as an upper bound
- Buildings and road segments need OpenStreetMap, which was not reachable when this result was built: shown as unknown, not zero
- This report points to where to look. Confirm important changes on the ground before acting.